

TPMS SW Library

Diagnostic of Pressure Sensor

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1. Scope

This document explains how the integrity of the pressure sensor is diagnosed thanks to the function `Check_IsBadPressure` that is available in the MLX91804/5 SW library with a revision 1.10.8 or higher.

2. Pressure Sensor Diagnostic

In MLX91804/5 the pressure sensor is a Wheatstone bridge. It includes 4 piezoresistive components depicted on the Figure 1 as 4 variable resistors. All 4 nodes of the bridge are bonded to the main die of the MLX91804/5. Two of them are used to supply the bridge, while another two (PS_SP and PS_SN) generate a signal of the applied pressure.

The main die has some controllable switchers that allow either to turn off the bridge to save power when the bridge is not used, or to switch it to different configurations used for tests.

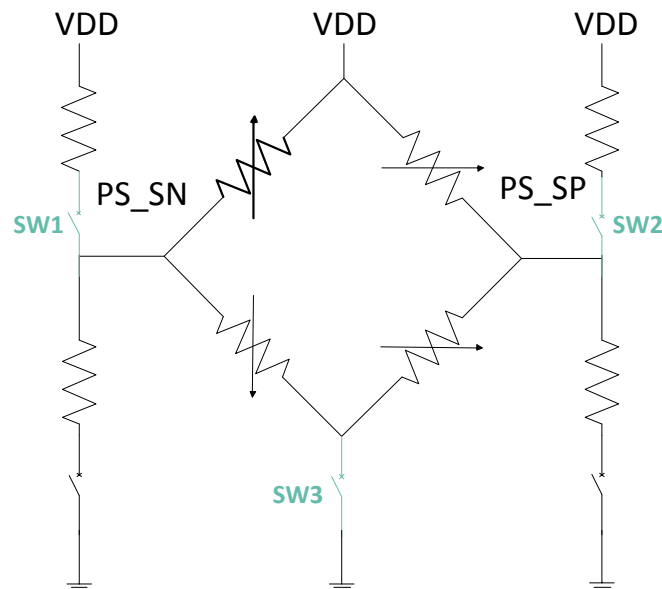


Figure 1: Pressure sensor in MLX91804/5

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To verify the pressure sensor, its bridge is configured by switchers SW1, SW2 and SW3 to have a predictable signal on the nodes PS_SN and PS_SP.

There are two different configurations (Config 1 and Config 2) that are used to check all of the four bonding wires connecting the pressure sensor with the main die.

The table below shows the switchers' state and the expected voltages on both PS_SN and PS_SP nodes for the operable pressure sensor. If one or more bonding wires become cut or short, the broken circuit won't result the expected voltage levels on the PS_SN and PS_SP nodes.

The real test results differ from the expected values due to the measurement noise, technological deviations and a signal drift over the temperature and VDD. The characterization done for the reference samples taken from the different lots was used to define the recommended test limits below:

Configuration	SW1	SW2	SW3	Voltage expected on the nodes PS_SN and PS_SP	Recommended test limits
Config 1	OPEN	OPEN	CLOSED	$\frac{VDD}{2}$	$MIN = 0.43 * VDD$ $MAX = 0.54 * VDD$
Config 2	CLOSED	CLOSED	CLOSED	$\frac{2 * VDD}{3}$	$MIN = 0.56 * VDD$ $MAX = 0.71 * VDD$

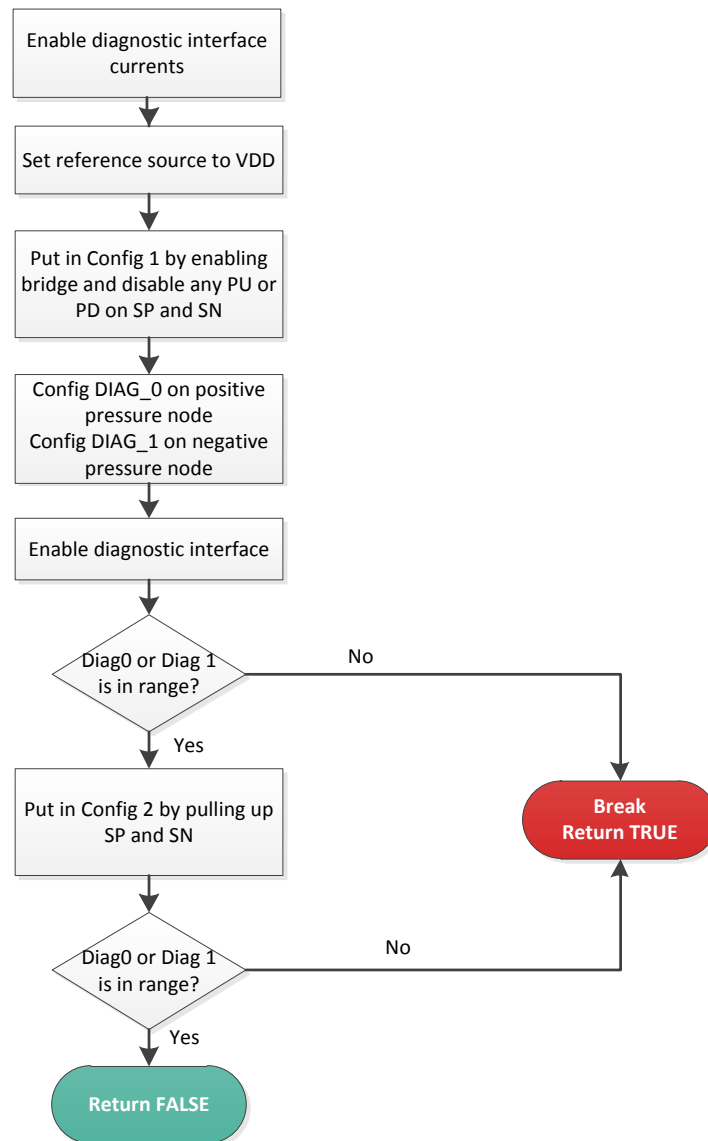
The characterization was done over the full VDD and temperature operating range.

The recommended values above determine the low and high thresholds in the library function Check_IsBadPressure.

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3. Firmware flow



4. Firmware example description

There is a firmware example of the pressure sensor diagnosis in the folder [examples/pressure_sensor_diagnostics](#) of the MLX91804/5 SW library.

The example includes simple **main.c** file that calls the library function **Check_IsBadPressure** recommended to verify the pressure sensor integrity. The function has no input parameters:

```
bool Check_IsBadPressure(void);
```

The function returns a Boolean value **“false”** if the pressure sensor is OK, and **“true”** if it is not OK.

The **main.c** file of the example calls the function **Check_IsBadPressure** in the endless loop and sets the output level of the DEBUG pin GPIO0 according to the state of the pressure sensor (logical “0” signals about a defective pressure sensor).